Product Troubleshooting Guide - Northstar Nexus

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Summary. Troubleshooting guide for Northstar Nexus covering error codes NX-1001, NX-2004, NX-3010, NX-4003 and NX-5000, required ticket information, version and upgrade compatibility, diagnostic checklists, worked examples and when to open or escalate a support ticket.

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1. Purpose and Scope

This guide helps Support Engineers, Customer Success staff and Engineering staff diagnose the most common problems customers meet with the Northstar Nexus Platform and Nexus Connect, and tells them what to ask the customer, what to check and what to do next. It also gives customers' technical contacts the same diagnostic logic, because Support often shares the resolution steps from this guide in knowledge base articles. It covers the five standard Nexus error codes, general diagnostic practice, version and upgrade compatibility, and the information that must be present in a ticket before Engineering is asked to investigate.

The guide applies to the Northstar Nexus Platform (workflow and integration platform with Workspaces, Pipelines, Connectors and Webhooks), Nexus Connect (API gateway and prebuilt connectors) and Nexus Insight (analytics add-on). It does not set support priorities, response targets or service credits, which are defined in CS-003 and the related service documents, and it does not describe how to escalate; for that see CS-001. API behaviours referred to here follow the API development standards in ENG-002. Knowledge base conventions are in CS-007.

2. Definitions

- **Workspace**: the tenant-level container in which a customer builds and runs Pipelines.
- **Pipeline**: a sequence of steps that moves and transforms data and calls Connectors.
- **Connector**: a prebuilt or custom integration with an external system, with its own version.
- **Webhook**: an HTTP callback that Nexus sends or receives to trigger or report on a Pipeline.
- **Tenant**: the customer's logical account in Nexus. Each tenant has a tenant ID.
- **Request ID**: a unique identifier returned with each API call and recorded in logs. It lets Engineering trace the exact request.
- **API token**: the credential a customer's system uses to call the Nexus API. Tokens are managed in Workspace settings.

- **Retry-After**: the HTTP header that tells a client how many seconds to wait before trying again after a rate limit response.
- **LTS**: Long-Term Support version, supported for 24 months. Non-LTS versions are supported for 12 months.

3. Roles and Responsibilities

- **Support Engineer (Level 1)** diagnoses using this guide, collects ticket information, applies documented resolutions, and decides whether Engineering must be engaged.
- **Support Lead** reviews cases that do not respond to the documented resolution, and decides on functional escalation to Engineering.
- **Engineering on-call and product engineers** investigate NX-5000 and any defect confirmed by Support; they request additional logs when the ticket lacks information.
- **Head of Support** owns this guide, reviews it every 12 months, and updates it when a new error code, a new version, or a recurring root cause is found.
- **Customer technical contact** provides the information requested, tests suggested changes, and confirms resolution.

4. Required Ticket Information

Every technical case in NorthAssist must include the following items before a ticket is passed to Engineering. A Support Engineer must ask the customer for any missing item at the first response, and must not delay an urgent P1 by waiting for completeness: for a P1 the engineer opens the Engineering request with whatever is known and continues collecting the rest in parallel.

Item	What it is	Why it matters
Tenant ID	The customer's tenant identifier	Locates the customer's data and configuration
Workspace name	The Workspace where the problem occurs	Narrows the Pipelines and Connectors involved
Nexus version	For example 4.2 LTS, 5.0 LTS or 5.1	Determines known issues and fixes
Request ID	Identifier returned with the failed call	Allows exact log tracing
Timestamp with timezone	The time of the failure, for example 2026-03-14 11:42 IST	Matches logs and rules out clock confusion
Steps to reproduce	Exact actions that cause the problem	Allows reproduction in a test environment
Impact statement	What business process is affected and how many users or transactions	Determines priority and urgency

A ticket without a request ID or a timestamp with timezone is the most common reason for Engineering returning a case to Support. Screenshots are helpful but never replace the request ID. Customer-specific data, such as personal data or secrets, must not be pasted into the ticket; ask the customer to share only identifiers and masked samples.

5. General Diagnostic Method

Follow the same sequence for every problem.

1 **Confirm the scope**: Is the issue in one Pipeline, one Workspace, one connector or the whole tenant? Is it a recent change or a long-standing behaviour?

2 **Check platform health**: Look at Northstar Status for an active incident or maintenance notice. If an incident is active, link the case to it and follow CS-001, section on Engineering incidents.
3 **Read the error code**: Match it to sections 6 to 10. Codes begin with NX- followed by four digits.
4 **Check what changed**: Token rotation, connector version updates, endpoint changes, new field mappings, volume increases or version upgrades.
5 **Test in isolation**: Ask the customer to run a single call or a minimal Pipeline with the same configuration.
6 **Apply the documented resolution**: Record what was done in the case.
7 **Verify**: Ask the customer to confirm the issue is gone and watch the Pipeline for at least one full run cycle.
8 **Decide on follow-up**: If the problem recurs, open a problem record and consider a knowledge base article under CS-007.

6. NX-1001 Authentication Failure

Meaning: The API call was rejected because the API token is expired or invalid. The platform returns HTTP 401.

Common causes:

- The token reached its expiry date.
- The token was revoked or regenerated, and a system still uses the old value.
- The token was copied with extra spaces or truncated.
- The call was sent to a different Workspace or region than the one the token belongs to.
- The service account that owns the token was deactivated.

Resolution:

1 Check in Workspace settings whether the token exists and its expiry date.
2 Rotate the token in Workspace settings and update the calling system with the new value.
3 Remove the old value from every place it was stored, such as configuration files and scheduler jobs.
4 Make a test call and confirm it succeeds.
5 Advise the customer to rotate tokens on a regular schedule and to store them in a secrets manager rather than in source code or shared documents.

Example: A customer's nightly Pipeline in a Pune-based retail chain started failing with NX-1001 on 1 April. The Support Engineer finds the token expired on 31 March. The customer rotates the token in Workspace settings, updates their scheduler, and the 2 April run completes. The Support Engineer records the cause as "token expiry" and suggests a calendar reminder 14 days before the next expiry.

Do not ask the customer to send the token value in a ticket. If a token is posted in a ticket or an email by mistake, the Support Engineer must tell the customer to rotate it immediately, remove the value from the case (internal note and attachments), and inform the Security Desk if the token was exposed beyond the customer's own staff.

7. NX-2004 Connector Timeout

Meaning: A Connector did not receive a response from the external endpoint within its timeout. The default connector timeout is 30 seconds. The timeout may be increased up to 120 seconds.

Common causes:

- The target system is slow, overloaded or under maintenance.
- The call returns very large payloads or runs a heavy query.
- Network path problems such as firewall rules, VPN or DNS issues between Nexus and the endpoint.
- The customer's endpoint rate-limits Nexus or requires an IP allow-list that is out of date.
- A recently changed endpoint URL or certificate.

Resolution:

1 Ask the customer to test the endpoint directly and measure its latency. If it regularly takes over 30 seconds, it is likely that the default timeout is too short for that call.
2 Check for recent changes to the customer's firewall, allow-list, certificates or URL.
3 Increase the Connector timeout in steps, for example from 30 to 60 seconds and then up to a maximum of 120 seconds, and re-test. Do not set 120 seconds by default; a long timeout hides slow endpoints and may block Pipeline capacity.
4 Reduce the payload size or split the request into smaller pages by using cursor pagination (default page size 50, maximum 200) when the Connector supports it.
5 If the endpoint is consistently slower than 120 seconds, advise the customer to change the integration pattern, for example to an asynchronous call with a Webhook callback.

Example: A Mumbai financial services customer sees NX-2004 on a Connector that calls its core banking system every 10 minutes. Direct tests show the response time is 45 seconds during month-end. The Support Engineer raises the timeout to 90 seconds, advises the customer to schedule the heavy export outside month-end processing, and logs the case. The error stops. The root cause is endpoint latency, not a Northstar defect, so no Engineering ticket is needed.

If NX-2004 appears across many customers at the same time, treat it as a possible platform issue, check Northstar Status, and contact the Engineering on-call through the Support Lead.

8. NX-3010 Rate Limit Exceeded

Meaning: The tenant sent more than 1,000 requests per minute, the default rate limit per tenant. The API returns HTTP 429 with a Retry-After header.

Common causes:

- A Pipeline or script that loops and calls the API without pauses.
- Several systems sharing one tenant and bursting at the same time, such as batch jobs starting at the top of the hour.
- A retry loop without backoff that multiplies requests after a failure.
- A sudden increase in business volume.

Resolution:

1 Ask the customer to read the Retry-After header and wait that many seconds before retrying.
2 Add exponential backoff with jitter to retries; never retry immediately in a tight loop.
3 Spread scheduled jobs across the hour rather than starting them all at the same minute.
4 Use batch endpoints, filters or webhooks instead of polling where possible.

5 If the customer legitimately needs a higher limit, record the business case, expected peak requests per minute and duration, and send a request to the Customer Success Manager and Product. Limit increases are a commercial and engineering decision and must not be promised by Support.

Example: A customer's inventory sync calls the API once per SKU. At 09:00 IST on a sale day, 4,000 SKUs are synchronised within a minute, and NX-3010 appears. The Support Engineer recommends batching 200 records per call, using the default page limit of 200 where it applies, and adding a 5-second delay between batches. The sync finishes in 3 minutes without errors.

9. NX-4003 Schema Mismatch in a Pipeline

Meaning: The data passing through a Pipeline step does not match the schema that the step expects. It often appears after a connector or source system changes.

Common causes:

- A source system added, removed or renamed a field.
- A Connector was updated to a newer version with a changed schema.
- The Pipeline's field mapping was not updated after a change.
- A data type difference, such as a number sent as text, or a date in a different format.
- A Pipeline built in one Nexus version was moved to another version without review.

Resolution:

1 Identify the step where the failure occurs from the Pipeline run details.
2 Compare the expected schema in the step with the actual payload (masked sample).
3 Re-map the fields in the Pipeline to the current schema.
4 Check the Connector version; if it changed recently, review the release notes and either pin the previous version temporarily or adopt the new schema.
5 Re-run the failed run for a small sample, then replay the backlog.
6 Add validation at the start of the Pipeline so that future schema changes are detected earlier and with a clearer message.

Example: A manufacturing customer's Pipeline fails with NX-4003 after its ERP team renamed the field "cust_code" to "customer_code". The Support Engineer asks for a masked payload, sees the new field name, and the customer re-maps the field. The customer then replays 312 failed records.

10. NX-5000 Internal Error

Meaning: An unexpected error occurred inside the platform. It is not caused by the request itself in a way the system can describe, so the customer cannot fix it alone.

Handling:

1 Collect the request ID, tenant ID, Workspace name, Nexus version, timestamp with timezone, steps to reproduce and impact statement.
2 Check Northstar Status for an incident.
3 Open an Engineering ticket at priority P2 or higher; do not leave NX-5000 as a P3 or P4 case unless Engineering has confirmed it is harmless. If production is down or data is at risk, set P1 and page the Engineering on-call.
4 Retry once after 5 minutes if the operation is safe to repeat; for operations that create resources, use the idempotency key so the retry does not create duplicates.

5 Give the customer a clear workaround if one exists, such as running the Pipeline manually or temporarily routing around the failing step.
6 Keep updating the customer at the update frequency of the priority, even when Engineering has no news.

Example: A customer's Webhook receiver reports NX-5000 for the last 20 minutes. The Support Engineer sees the request ID in the response, checks Northstar Status, finds nothing, and opens a P2 Engineering ticket with the details. Engineering finds a defect in the connector's retry logic. The workaround, disabling the retry setting for the Workspace, is shared within 90 minutes.

11. Quick Reference

Code	Meaning	First action	Limits to remember
NX-1001	Authentication failure	Check and rotate the API token in Workspace settings	Token expired or invalid
NX-2004	Connector timeout	Test endpoint latency, then raise timeout	Default 30 seconds, up to 120 seconds
NX-3010	Rate limit exceeded	Wait for Retry-After and add backoff	1,000 requests per minute per tenant
NX-4003	Schema mismatch	Re-map fields, check connector version	Compare expected and actual schema
NX-5000	Internal error	Collect request ID, open P2 or higher ticket	Engineering must investigate

12. Versions and Upgrade Compatibility

Many apparent defects are due to version differences. Before diagnosing, always confirm the customer's Nexus version and whether it is still supported.

Version	Release date	Type	Support status
4.2 LTS	2025-01-20	LTS	Supported until 2027-01-20
4.3	2025-06-03	Non-LTS	Support ended 2026-06-03
5.0 LTS	2026-02-10	LTS	Supported until 2028-02-10
5.1	2026-08-18	Non-LTS	Supported until 2027-08-18

LTS versions are supported for 24 months and non-LTS versions for 12 months. A customer on 4.3 is on a version whose support ended on 2026-06-03; the Support Engineer should still help with diagnosis but must explain that a fix may require an upgrade. The supported upgrade paths are:

- 4.2 to 5.0 directly.
- 4.3 must go through 5.0 or 5.1, and 4.3 to 5.1 is a supported direct upgrade.

Upgrade planning is the responsibility of the customer with the CSM and Implementation team. Support must not tell customers that an unsupported path will work. If a customer asks about a

different path, for example 4.2 to 5.1, Support must check with Product before replying, because only the paths above are confirmed.

After any upgrade, new errors, especially NX-4003 and NX-2004, are more likely because connector versions, schemas and defaults may change. Ask the customer whether an upgrade occurred in the last 14 days.

13. API Behaviour Reminders

The following behaviours, aligned with ENG-002, often explain customer questions.

- **Versioning**: URLs include a version such as /v1/. Breaking changes appear only in a new major version, and public API deprecations come with at least 12 months notice and a Sunset header.
- **Pagination**: cursor pagination with a default page size of 50 and a maximum of 200.
- **Rate limits**: 1,000 requests per minute per tenant by default; excess requests receive HTTP 429 with Retry-After.
- **Timeout**: the default request timeout is 30 seconds.
- **Errors**: responses use the problem-details format, which includes a type, a title, a status and a detail field. Customers should log the whole error body, including the request ID.
- **Authentication**: OAuth 2.0 client credentials for service access.
- **Idempotency**: POST calls that create resources must carry an idempotency key, so a retry does not produce a duplicate.

14. When to Escalate to Engineering

A Support Engineer must open an Engineering ticket, after recording the information in section 4, when any of the following is true.

- The error is NX-5000.
- The documented resolution has been applied and the problem persists.
- The problem affects several customers or several Workspaces at the same time.
- A defect is suspected, such as incorrect output from a Connector with correct input.
- There is any sign of data loss, corruption or exposure; in this case page the Engineering on-call, notify the Security Desk and follow CS-001.

The Support Lead decides on priority and functional escalation. Escalation levels and timing are in CS-001, and no delivery date for a fix may be given to the customer without Engineering confirmation.

15. Knowledge Base and Continuous Improvement

When a case reveals a new cause or a better resolution, the Support Engineer should propose a knowledge base article. Articles follow the template Problem, Environment, Cause, Resolution and Related articles, are reviewed by two people, are reviewed every 12 months, and move through the statuses Draft, Published and Archived. They must not contain customer-specific data. The Head of Support reviews the top 10 error causes each quarter with Product and Engineering to decide on product fixes, clearer error messages or documentation changes.

16. Related Documents

- CS-007 for knowledge base article standards.

- CS-006 for the product release and compatibility information that complements this guide.
- ENG-002 for the API development standards behind the behaviours in section 13.
- CS-001 for escalation levels and CS-003 for priorities and response targets.

17. Revision History

Version	Date	Change
1.0	2025-05-08	First release of the Nexus troubleshooting guide
1.1	2025-11-26	Added NX-5000 handling, upgrade paths and API behaviour reminders